Calculus Formula Sheet

Mathematics for Science, Week 6

Numbers, Sets and Powers

We write $\mathbb{N}$ for the natural numbers, $\mathbb{Z}$ for the integers and $\mathbb{R}$ for the real numbers, so that $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{R}$. An interval such as $[a, b] = \{x \in \mathbb{R} \mid a \le x \le b\}$ contains its end points, while (a, b) does not.

For $a > 0$ and any real numbers m and n,

$$a^m a^n = a^{m+n}, \qquad (a^m)^n = a^{mn}, \qquad \sqrt[n]{a} = a^{1/n}, \qquad a^{-n} = \frac{1}{a^n}.$$

Limits and Derivatives

With the Greek letter θ for an angle, two limits appear again and again:

$$\lim_{\theta \to 0} \frac{\sin \theta}{\theta} = 1 \qquad \text{and} \qquad \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n = e.$$

If $f(x) = x^n$ then $f'(x) = nx^{n-1}$. The quotient rule says

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}, \qquad v \neq 0.$$

Integrals

For $n \neq -1$,

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C, \qquad \int_0^{\pi} \sin\theta \, d\theta = 2, \qquad \int_{-\infty}^{\infty} e^{-x^2} \, dx = \sqrt{\pi}.$$

Sums and Series

The sum of the first n natural numbers and the geometric series with ratio $|r| < 1$ are

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2} \qquad \text{and} \qquad \sum_{k=0}^{\infty} r^k = 1 + r + r^2 + \cdots = \frac{1}{1-r}.$$

In a sentence a sum is kept small, $\sum_{k=1}^{n} k$, and $\displaystyle\sum_{k=1}^{n} k$ forces display size. The binomial theorem expands a power of a sum:

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k.$$